at most one route: we must wait it out—second must follow second as tick by tock now methodically gives way to then. And this assumes that "then" is later than "now." If then precedes now, experience dictates that there is no route at all; traveling to the past seems not to be an option. Unlike travels through space, travels through time appear to be anything but a matter of choice and whim. When it comes to time, we get dragged along in one direction, whether we like it or not.

Were we able to navigate time as easily as we navigate space, our worldview would not just change, it would undergo the single most dramatic shift in the history of our species. In light of such undeniable impact, I am often struck by how few people realize that the theoretical underpinnings for one kind of time travel—time travel to the future—have been in place since early last century.

When Einstein discovered the nature of special relativistic spacetime, he laid out a blueprint for fast-forwarding to the future. If you want to see what's happening on planet earth 1,000, or 10,000, or 10 million years in the future, the laws of Einsteinian physics tell you how to go about it. You build a vehicle whose speed can reach, say 99.9999999996 percent of light speed. At full throttle, you head off into deep space for a day, or ten days, or a little over twenty-seven years according to your ship's clock, then abruptly turn around and head back to earth, again at full throttle. On your return, 1,000, or 10,000, or 10 million years of earth time *will* have elapsed. This is an undisputed and experimentally verified prediction of special relativity; it is an example of the slowing of time with the increasing of speed described in Chapter 3.[8] Of course, since vehicles of such speed are beyond what we can build, no one has tested these predictions literally. But as we discussed earlier, researchers have confirmed the predicted slowing of time for a commercial airliner, traveling at a small fraction of light speed, as well as that of elementary particles like muons racing through accelerators at very nearly the speed of light (stationary muons decay into other particles in about two millionths of a second, but the faster they travel the slower their internal clock's tick, and so the longer the muons appear to live). There is every reason to believe, and no reason not to believe, that special relativity is correct, and its strategy for reaching the future would work as predicted. Technology, not physics, keeps each of us tethered to this epoch.[42]